

GUSYNTH

POLYPHONIC GUITAR SYNTHESIZER PEDAL

Authors

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Group 2 Logo ?

UC SANTA CRUZ



Design Statement

Need: Guitarists want to create the sound of another instrument without removing their guitar during live performance or learning a new instrument.

Goal: Allow guitarists to play the sounds of different instruments with their guitar in real time — with no perceptible delay.

Design Objectives

- Polyphonic Note Detection - Accurately infer the number of distinct notes played in a chord and their individual pitches.
- Low Latency (<10 ms) - The player must not perceive any delay between playing and hearing the synthesized sound.
- Pedal Form Factor - Fits a standard pedalboard, accepts 9V DC center-negative power, and matches the design language of existing guitar effects pedals.

Personas

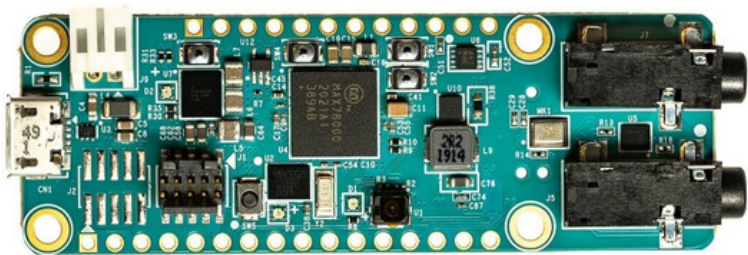
Bedroom Guitarist - Casual hobbyist expanding their creative palette. Wants expressive sounds from gear they already own with easy setup and no learning curve.

Recording Studio- Producer or session musician needing precise, repeatable tones. Values low noise and consistent output.

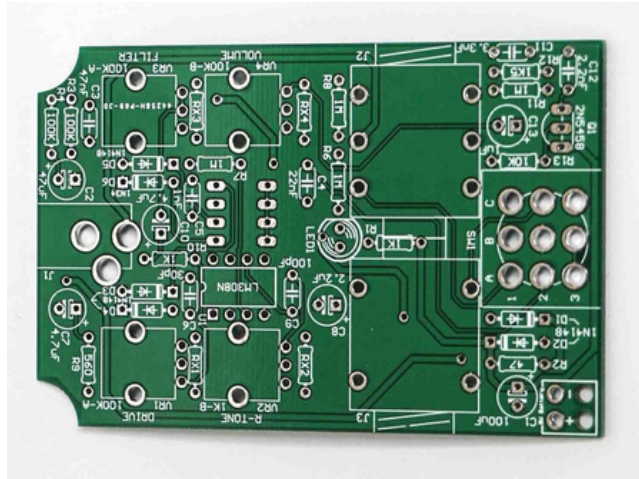
Live Performer - Performing musician who needs reliable, low-latency response on stage. Prioritizes fast controls, and a sound that cuts through a live mix.

Hardware Desgin

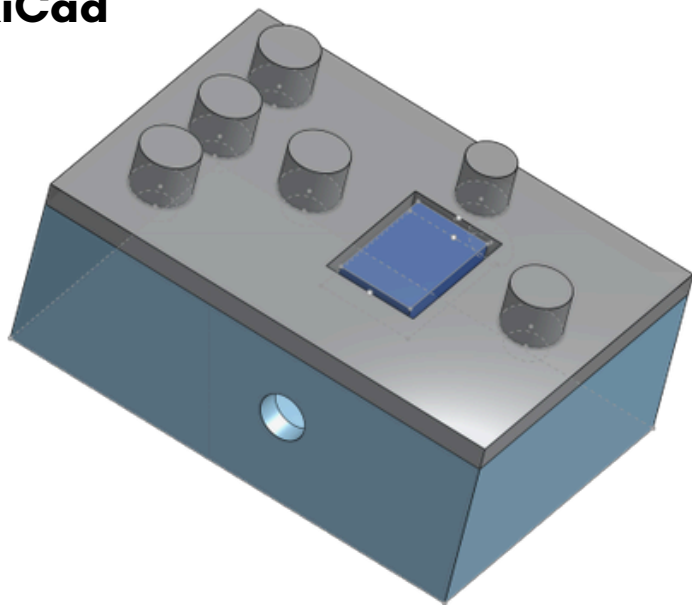
MAX78000FTHR MCU



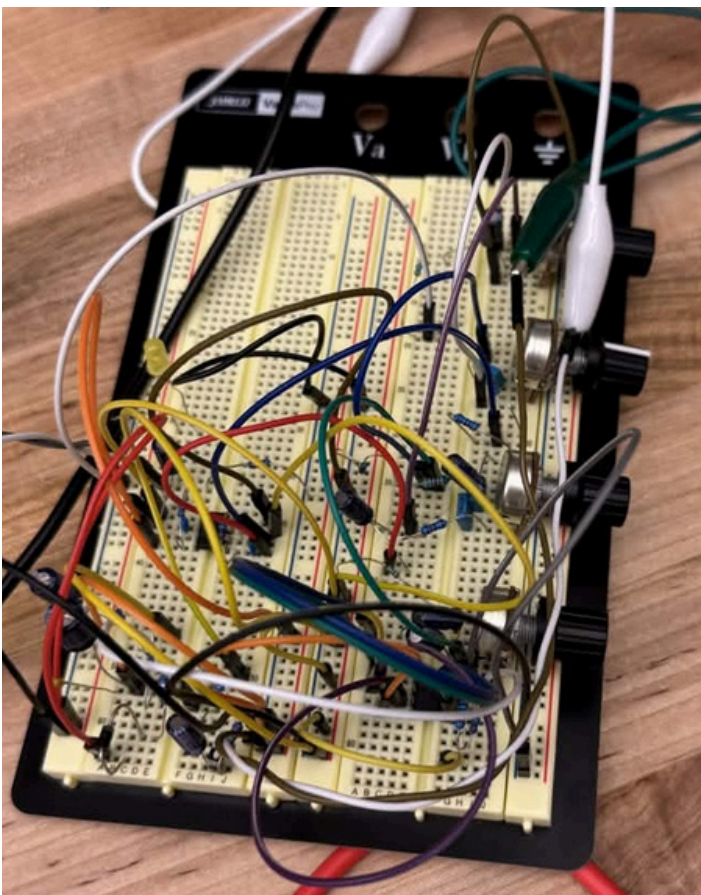
PCB



KiCad



Overdrive/Fuzz Circuit

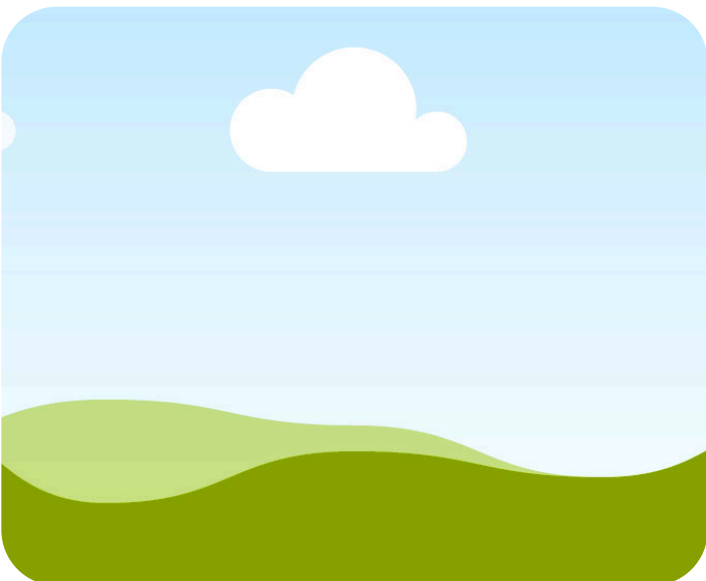


Pots/Knobs

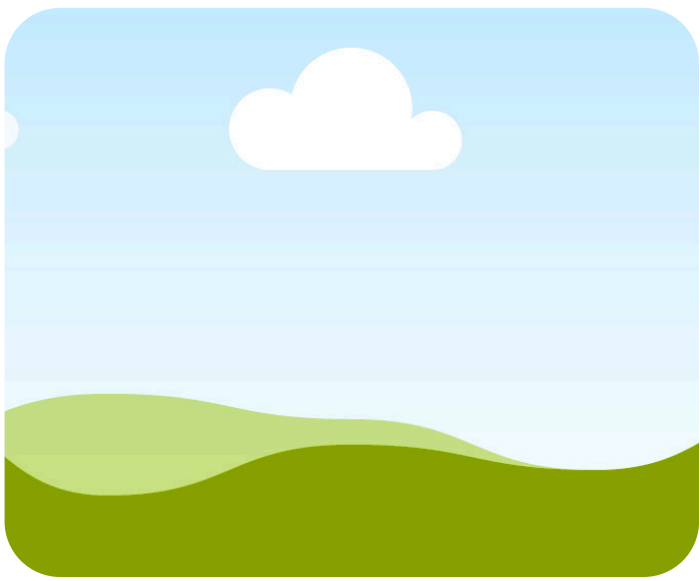


Software Design

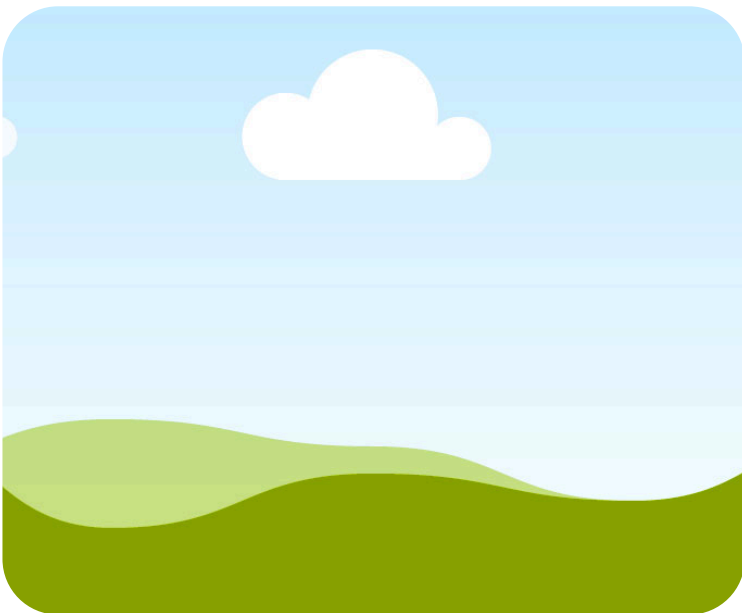
NN Training Curves



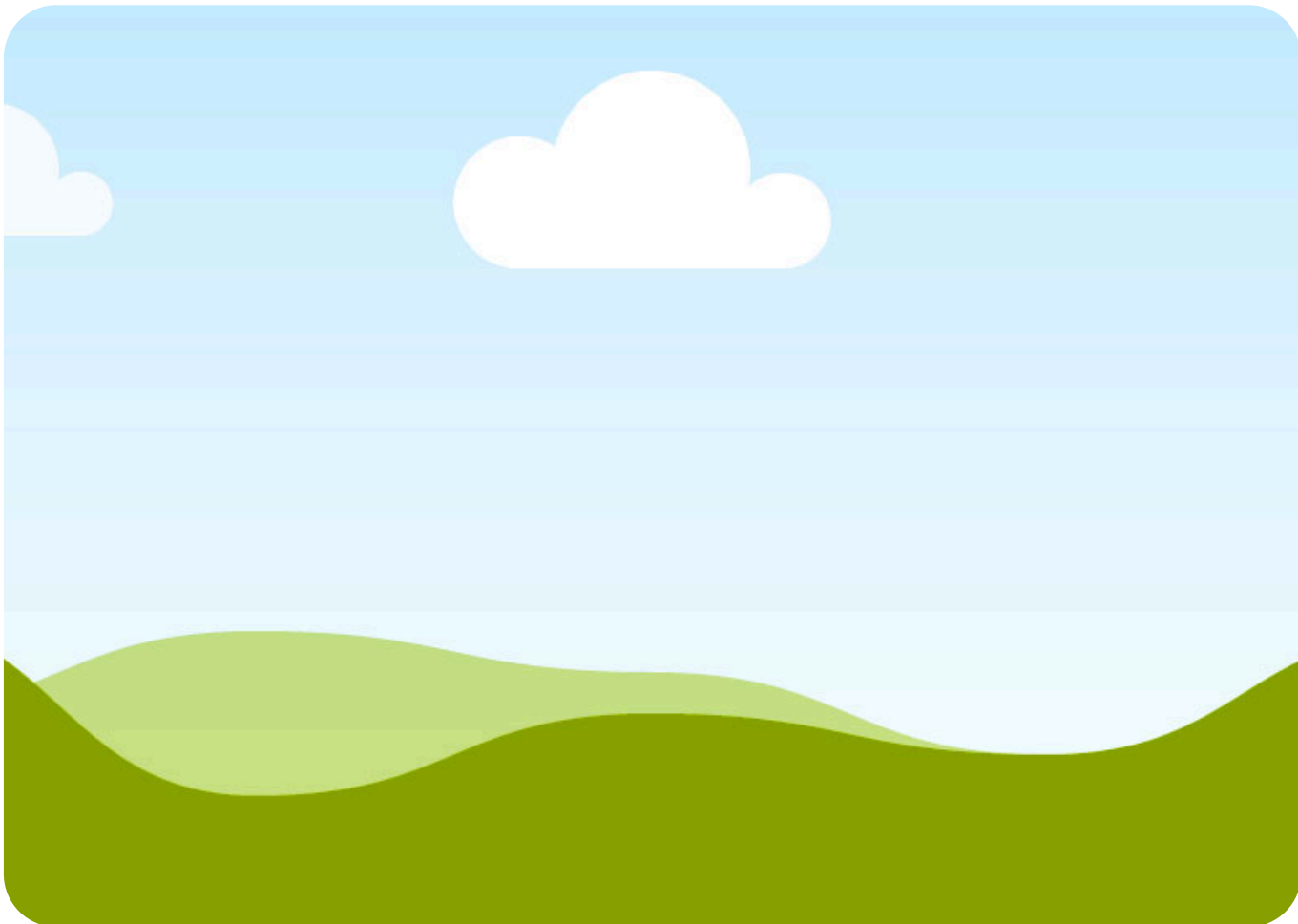
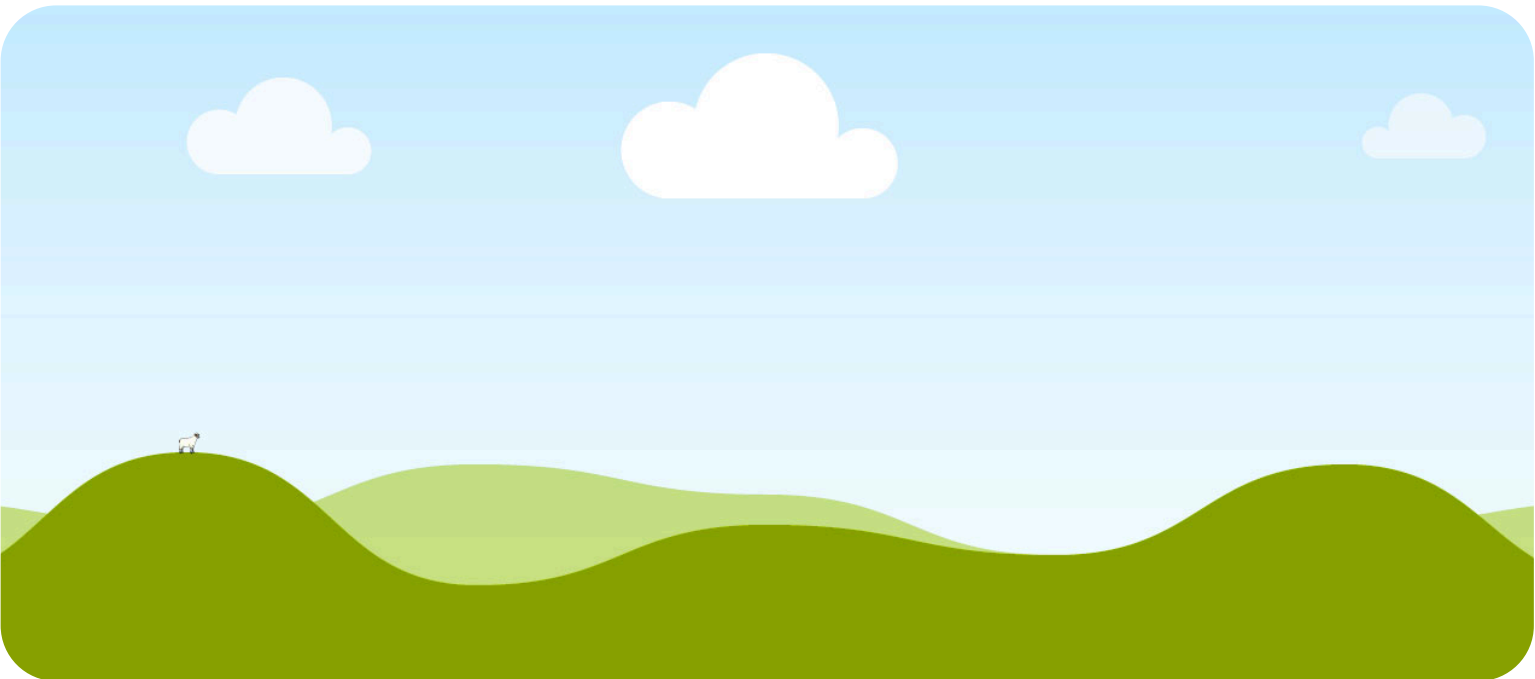
NN Architecture



Output/Visualizations



Functional Prototype



Key Sources & Acknowledgements